

9/7/26

* Create a structure variable of a student estimate the size and structure manually. then.
=> write a c program to find the size of structure now verify size of operators. now observe both the outputs. Are they your answer

=> C programming

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
int id;
```

```
char grade;
```

```
float marks;
```

```
};
```

```
int main ()
```

```
{
```

```
struct student s;
```

Part 4 (Size of structure student = 12 bytes / 16 bytes, size of 15))
 returns;

Manual size calculation

<u>Data member</u>	<u>Datatype</u>
id	int
grade	char
Marks	float
Total	

<u>Size</u>
4 bytes
1 byte
4 bytes
12 bytes

Actual size using size of structure student = 12 bytes

Why are they different?

The Outputs are different

- Manual size = 12 bytes (4+1+4)
- size of result = 12 bytes (on most compiler)

This difference occurs because the compiler adds padding bytes b/w member align data in memory for faster access

Memory layout

<u>member</u>	<u>size</u>	<u>offset</u>
id	4 bytes	0 or 3
grade	1 byte	4
padding	3 bytes	5-7
Marks	4 bytes	8-11